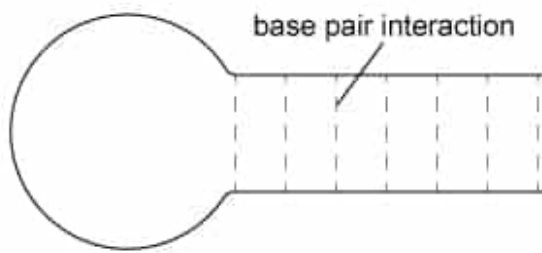
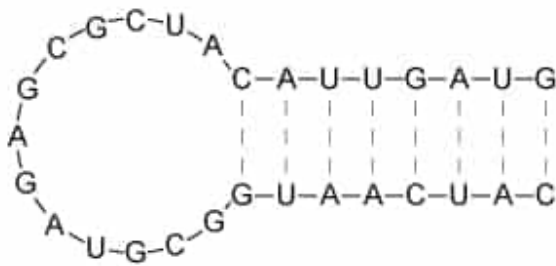


Certain RNA molecules can fold into structures known as hairpins, as depicted below.



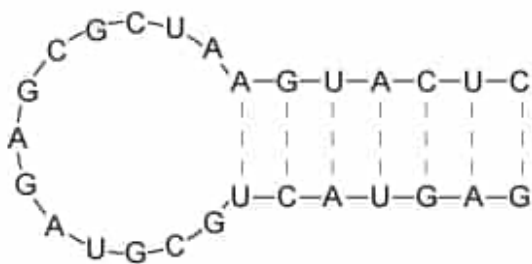
Which of the following RNA hairpins is the most stable?

☐ A.



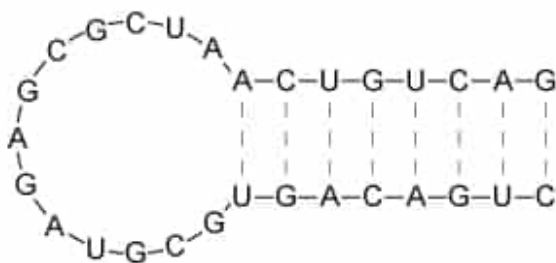
[6%]

☐ B.



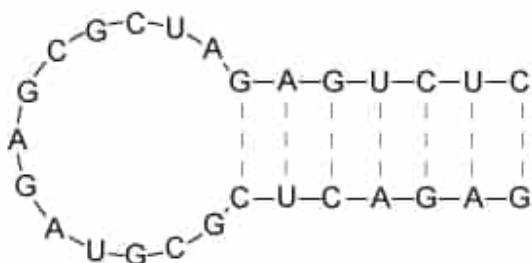
[2%]

☒ C.



[37%]

☐ D.



[54%]

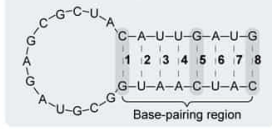
Correct

37% answered correctly

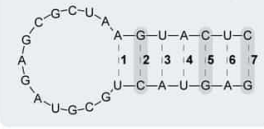
Explanation:

Nucleic acid structures with more total base pairs
& more G-C pairs are more stable

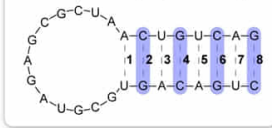
Choice A: 8 total base pairs, 3 G-C base pairs



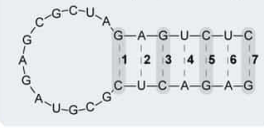
Choice B: 7 total base pairs, 3 G-C base pairs



Choice C: 8 total base pairs, 4 G-C base pairs



Choice D: 7 total base pairs, 4 G-C base pairs



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Nucleic acids **base-pair** with each other according to [Watson-Crick](#) hydrogen bonding rules. RNA is typically single-stranded and may wrap around on itself to form base pairs between nucleotides within the same strand. For example, RNA commonly forms structures known as hairpins, in which a loop with no base pairing is adjacent to a stretch of several base pairs.

Structures such as hairpins are stabilized by hydrogen bonds, and the more hydrogen bonds that are present, the more stable the structure. In RNA, base pairing occurs between adenine (A) and uracil (U), or between cytosine (C) and guanine (G). The structures of A and U facilitate the formation of **2 hydrogen bonds in A-U base pairs** whereas **G-C pairs form 3 hydrogen bonds** with each other. Because G-C base pairs contain more hydrogen bonds than A-U base pairs, they are **stronger and more stable**.

For RNA hairpins that have the same number of base pairs, the hairpin that contains the *most* G-C pairs will be the most stable. For hairpins that have the same number of G-C pairs, the one with the *most* total base pairs will be the most stable. **Choice C** shows a hairpin that contains 8 total base pairs, 4 of which are G-C pairs. Accordingly, it is the most stable of the options given.

(Choice A) This hairpin has 8 total base pairs, 3 of which are G-C pairs. It has fewer G-C pairs than Choice C, so it is *less* stable.

(Choice B) This hairpin contains 7 total base pairs and 3 G-C pairs. It is the *least* stable of the options shown.

(Choice D) This hairpin has 4 G-C pairs, but only 7 total base pairs. It has fewer total pairs than Choice C, so it is *less* stable.

Educational objective:

RNA structures such as hairpin loops are stabilized by hydrogen bonds between base pairs. G-C base pairs form more hydrogen bonds than A-U base pairs. Therefore, structures with more total base pairs and more G-C base pairs are more stable.

Time spent:0 sec

Subject:
Biochemistry

QID:402693

System:
Carbohydrates, Nucleotides, and Lipids